KOI FALSE-POSITIVE PROJECT · MIND MAP · TEACHER PLANNING MAP

Can a computer spot a fake planet?

The same map with session timing, teaching notes, deliverables and marking.

STAGE A
WEEK 1 · Notebook Part 1

The catalogue, and the clue in it

- Download the KOI table from the NASA Exoplanet Archive. Keep CONFIRMED and FALSE POSITIVE, drop CANDIDATE.
- Plot depth, period and impact parameter before building anything.
- Work out the guessing baseline: about 47% are false positives, so accuracy on its own means very little.
- **Run the leakage check and set aside nine columns that were filled in after somebody knew the answer.**
- Train a Random Forest on the eight columns that are left, splitting by star.
- **Put koi_score back on purpose. Accuracy jumps to 1.00.**
- **A very high score is a reason to look for a mistake.**

TEACHING NOTES

Session 1, 60-75 min. Ask the class what accuracy a program gets if it always answers 'false positive', before telling them.

Let students say out loud that the leaky model just copied the answer. Do not rush this moment.

kepler_name is the subtle one: only confirmed planets have one, so an empty field is the label.

STAGE B
WEEK 1 · Notebook Part 2

From brightness to numbers

- Take 50 confirmed planets and 50 false positives with a fixed random seed, and download with author="Kepler" so all come from one pipeline.
- **Clip spikes upward only. The symmetric setting would delete deep eclipses.**
- **Detrend with flatten(mask=...) so the fit ignores the in-transit points.**
- Measure in three windows: core 0.30, full 0.50, baseline 1 to 4 x duration.
- **Test the functions on a transit you injected yourself before trusting real data.**
- Run the loop. It saves progress every ten targets, so it can be restarted.

THE TEN FEATURES (* = a student TODO)

```
depth_ppm duration_hr n_transits scatter_oot_ppm snr v_shape
*odd_even_sigma *secondary_ppm depth_scatter *rho_star_cgs
```

TEACHING NOTES

Session 2, then the download loop as homework: 20-40 min, a few gigabytes. Keep a pre-measured CSV in reserve in case downloads fail.

Demonstrate flatten() once WITHOUT the mask so the class sees the depth shrink.

Injected test should give: planet v_shape 1.07, secondary 0; binary 1.42, 1500 ppm.

THE BIG IDEA

A light curve records how much light was lost, and when.

It never records where on the sky the light was lost.

So a pair of eclipsing stars near your target makes an identical dip.

That kind of false positive can never be found from brightness alone.

Measuring where that limit sits IS the scientific result.

THE QUESTION

Can a computer spot a fake planet?

What fraction of Kepler's false positives can be identified from the shape of the light curve alone?

WHAT TO COLLECT

STAGE A
A written note: the leaky model's accuracy, the column it relied on, and the list of banned columns with a reason for each.

STAGE B
measured_features.csv with 80+ objects, four folded plots with the windows drawn on, and a count of the downloads that failed.

STAGE C
Classification report, confusion matrix and precision-recall curve from the student's own features, plus a comparison with Stage A.

STAGE D
The flag comparison chart and a written conclusion: a number, an uncertainty, a baseline comparison and a physical reason.

MARKING
Weight Method, Honesty and Conclusion above the recall achieved. Check the split is grouped, the metric suits the imbalance, and the features were tested on an injected signal first.

STAGE C
WEEK 2 · Notebook Part 3

Training the model

- From here on every number came from the brightness data. No catalogue columns are used.
- **Split by kepid, never by row, so one star cannot sit on both sides.**
- **Read the recall for the false-positive class first. That is the number this project is about.**
- Draw the precision-recall curve with the guessing line marked, not a ROC curve.
- Repeat with grouped 5-fold cross-validation for a fairer score.
- Rank the feature importances, then plot the two strongest against each other.

TEACHING NOTES

Session 3, 60-75 min, after finishing the three TODO features.

Students who used train_test_split will trip the assertion. That failure is the intended teaching moment.

Watch for over-claiming on the scatter plot. The two clouds overlap; ask for an estimate of the overlap fraction.

Do not expect Stage A to be beaten. Explaining the gap is worth more.

STAGE D
WEEK 2 · Notebook Part 4

Finding the limit

- Attach the four koi_fpflag_* columns. Analysis only, never training.
- Split the true false positives into the ones caught and the ones missed.
- Work out what fraction of each group carries each flag.
- **Check whether the centroid-offset flag dominates the objects you missed.**
- **Centroid offset is measured from the camera pixels, not the light curve, so no feature of yours could have caught it.**
- Write the answer as a number with an uncertainty, compared with guessing, and with a physical reason.

TEACHING NOTES

Session 4, 60-75 min. This is the assessment centrepiece.

If fewer than three objects were missed, pool the cross-validated predictions instead of using one split.

Mark the conclusion on its structure, not on the size of the recall.

0.62 recall with an honest analysis beats 0.98 with a leaked column.